

GAVIN MASON

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TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Java, SQL, Bash, C, R, HTML/CSS

Machine Learning: PyTorch, TensorFlow, scikit-learn, LSTM, deep learning, MLOps, model deployment, synthetic data generation

AI/LLM & Data: LangChain, LangGraph, vLLM, ChromaDB, RAG, vision models, pandas, NumPy, ETL pipelines, web scraping

Frontend: React, Vue 3, Nuxt 3, TypeScript, TailwindCSS, Vite, GraphQL, Jest, Playwright, Cypress

Backend: Node.js, Python, FastAPI, Spring Boot, Strapi.js, PostgreSQL, REST APIs, microservices

Cloud/DevOps: AWS (EC2, Bedrock, S3), Kubernetes, Docker, Kong AI Gateway, ArgoCD, CI/CD, Linux, Git/GitHub

EXPERIENCE

Software Engineer

MITRE Corporation | Summer 2024 – Present | San Antonio, TX

- Lead and sole engineer architecting production machine learning threat detection system processing 100K+ daily API requests. Built LSTM-based classifier achieving 95%+ accuracy in identifying Cost Harvesting attacks with <100ms inference latency
- Engineered comprehensive end-to-end synthetic data pipeline generating 1,000,000+ Kong Gateway logs with mathematically validated attack patterns and temporal burst detection algorithms, reducing real-world data collection costs by 90%
- Deployed production-ready FastAPI microservice on AWS EC2 with systemd orchestration, implementing automated feature engineering pipeline using pandas and NumPy. Built comprehensive monitoring tracking model performance, API latency, and real-time threat classification metrics
- Led enterprise Cybersecurity AI R&D as primary engineer on confidential initiatives, delivering production-ready systems and prototypes. Authored and presented technical architecture briefs to C-suite leadership including CTO, directly influencing security infrastructure investment decisions and organizational AI strategy
- Architected and deployed full-stack RAG chatbot with React frontend and Python backend, leveraging LangChain and LangGraph for agentic reasoning workflows and ChromaDB vector database for semantic search. Implemented multi-format document ingestion pipeline using Docling, integrated vision models for diagram extraction, and deployed on Kubernetes with Kong API Gateway routing
- Piloted enterprise deployment of secure 7B parameter LLM coding assistant using vLLM for optimized inference, Kong AI Gateway for security policy enforcement, and Kubernetes for horizontal scalability. Partnered with security vendors to integrate enterprise-grade controls, improving engineering team productivity while maintaining strict security compliance

Software Engineering Intern

MITRE Corporation | Summer 2022 – Summer 2024 | Tampa / Orlando, FL

- Developed custom Grafana plugins for enterprise supply chain vulnerability visualization, enabling real-time tracking and analysis of CVE data across microservices. Significantly reduced vulnerability identification time and enabled security teams to prioritize remediation efforts based on severity and exploitability metrics
- Enhanced Java Spring Boot microservice platform and led comprehensive research into SBOM comparison tool evaluation. Strengthened CI/CD pipelines with automated dependency scanning and vulnerability detection, driving organization-wide adoption of supply chain security best practices across engineering teams
- Spearheaded full-stack modernization of saf.mitre.org using Nuxt 3, TailwindCSS, Strapi.js, and PostgreSQL backend. Implemented GraphQL APIs for efficient data fetching and comprehensive Playwright E2E test suite, creating a unified platform for MITRE's Security Automation Framework serving the broader cybersecurity community

PROJECTS

SEE Lunar Base Architecture (NASA / Florida Space Institute) | Nuxt 3, Vue 3, HLA, Distributed Systems

- Built full-stack distributed simulation platform for NASA's SEE Smackdown competition supporting multiple concurrent teams. Integrated Vue 3 web interface for synchronized lunar habitat simulations supporting NASA Artemis mission and research
- Developed High-Level Architecture simulation federate with 3D lunar base models and real-time state synchronization across distributed systems. Implemented Trick HLA and Distributed Object Network (DON) infrastructure enabling complex multi-system experiments adhering to NASA simulation standards

EDUCATION

Bachelor of Science in Computer Science, University of Central Florida

Graduated: Spring 2024 | GPA: 3.75 | Orlando, FL | Campus Involvement: Knight Hacks @ UCF